
[WORKING TITLE: Certified Access: Test-Time Compute on a Conservative Memory]

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Abstract

1 Test-time compute mechanisms, such as retries, confidence-based escalation, and
2 non-local routing, are conventionally deployed as black-box augmentations to
3 trained models. These approaches typically introduce additional forward passes
4 or learned halting heads without providing a structural account of the specific
5 capabilities acquired by the additional compute, nor do they quantify the costs
6 incurred against the model’s fundamental stability guarantees. In this work, we
7 propose a theoretical and empirical framework for certified access in causal, sym-
8 plectic associative memories. When the underlying latent state (q, p) is advanced
9 by a damped symplectic (velocity-Verlet) step of a learned Hamiltonian, every
10 compute mechanism that expands access has an exact phase-volume Jacobian, a
11 bounded energy change, and an exact latch-transport law. We formalize latent
12 reachability into two distinct failure modes. First, reach failures occur when a tar-
13 get lies outside a kinematic causal box C_T . Second, escape failures occur within
14 C_T when a state is trapped behind latent surface barrier. To resolve escape fail-
15 ures, we deploy a Lorentz squeeze, which injects bounded, re-absorbable energy
16 that expands reach at test time. To resolve reach failures, we introduce a gated
17 canonical translation (a one-step channel across C_T), which expands latent reach
18 at $\det J = 1$ exactly. We verify this certificate stack on a designed analytic testbed,
19 and show that reaching just beyond C_T with the squeeze requires energy growing
20 quadratically in the excess distance covered, whereas the gated channel’s energy
21 requirement is fixed and independent of distance. Furthermore, we separate trans-
22 port from latent state-replacement, demonstrating that an untrained state-replacing
23 map annihilates phase volume and erases stored charge, while the gated channel
24 preserves it. Finally, we situate these mechanisms on a learned physics-structured
25 memory, demonstrating memory-agnostic calibrated compute-rationing and eval-
26 uating scaling boundaries for non-local routing against a physics-free baseline.
27 The primary contribution of this work lies in the theoretical formalization and em-
28 pirical verification of the certificate stack, prioritizing structural discipline over
29 isolated benchmark performance.

1 Introduction

31 Current deep learning architectures increasingly rely on adaptive test-time compute to dynamically
32 allocate inference budgets. Approaches such as learned halting (ACT, Graves, 2016; PonderNet,
33 Banino et al., 2021), confidence-gated early exits (CALM, Schuster et al., 2022), learned routing
34 (Mixture-of-Depths, Raposo et al., 2024; Mixture-of-Experts, Shazeer et al., 2017), and energy-
35 based verification (Gladstone et al., 2025) have established the utility of input-conditional escalation.

36 However, these mechanisms generally bolt a learned scalar gate onto a feedforward stack. Conse-
 37 quently, escalating compute merely replicates the standard forward operation without providing a
 38 structural guarantee of how the additional computation explicitly alters the latent state of the model.

39 To resolve this limitation, we introduce a framework that situates test-time compute within the
 40 physics of conservative dynamical systems. If the queried memory is a conservative latent state
 41 (q, p) advanced by a structure-preserving symplectic integrator of a learned separable Hamiltonian
 42 $H(q, p) = T(p) + V_\theta(q)$, test-time compute manifests as access to a defined phase space. Within
 43 this regime, the physical properties of the phase space strictly define every access mechanism. We
 44 formalize this paradigm as certified access, where each mechanism is accompanied by a physical
 45 certificate. This certificate comprises the Jacobian determinant $\det J$ to certify phase volume conser-
 46 vation, a bounded energy change to quantify injected energy, and a latch-transport law to determine
 47 the transformation of content stored in protected continuous-symmetry directions. Crucially, these
 48 certificates are verifiable theorems derived from the underlying symplectic structure rather than em-
 49 pirical curves fit to specific datasets. We state each certificate as a theorem and verify it on a designed
 50 testbed: the exactness claims ($\det J$, the energy change and the latch-transport law) hold to machine
 51 precision, while the re-absorption and channel-hop thresholds are verified to leading order.

52 **Reference architecture and terminology.** Our reference memory is the Causal Learning Unit
 53 (CLU), introduced as CHLU in Jawahar and Pierini (2026). The empirical results in this work as-
 54 sume the reference training objective of the CLU but evaluate generalized properties of the learned
 55 Hamiltonian. Throughout this work, the learned inertial mass M_i , which dictates the causal box
 56 C_T , is distinct from the spectral mass μ_k , which defines normal-mode stiffness or loosely the cur-
 57 vature of the latent surface. Reachability mechanisms are parameterized by M , whereas retention
 58 mechanisms not discussed in this work are parameterized by μ .

59 Contributions.

- 60 1. **Formalization of failure modes:** We partition reachability failures into kinematic reach,
 61 where the target lies outside a causal box C_T bounded by $L_i = T\epsilon c/\sqrt{M_i}$, and energetic
 62 escape, where the target is inside C_T but shielded by a potential barrier. These are resolved
 63 by distinct mechanisms carrying different certificates.
- 64 2. **Verification of the certificate stack:** We verify our theoretical predictions on an analytic
 65 testbed using dimensions 2 and 4, 5 seeds, and oracle placement. We confirm that the
 66 Lorentz squeeze cures escape with bounded injection and extends reach at an energy expo-
 67 nential in the rapidity. We also verify that the wormhole mechanism cures reach continu-
 68 ously with $\det J = 1$, a fixed energy change exact to zero, and exact latch transport.
- 69 3. **Empirical measurement of latch transport:** We demonstrate that an untrained state-
 70 replacing map has $\det J = 0$ exactly, resulting in the complete erasure of stored Goldstone
 71 charge. In contrast, the certified wormhole transports the charge with preserved spread.
 72 We explicitly establish that phase volume alone is insufficient for this transport; a matched
 73 channel is required. Furthermore, a bounded energy change is necessary but insufficient for
 74 Bounded-Input Bounded-Output (BIBO) stability without coercive-component screening.
- 75 4. **Memory-agnostic calibrated compute-rationing:** We implement a self-calibrating
 76 compute-rationing gate on an escalatable learned memory. We observe a $4.81 \pm 0.44\times$
 77 reduction in relaxation steps at the kv16 capacity level at a 400-epoch budget. The primary
 78 asset is escalatability and rationing rather than baseline energy signal superiority, and the
 79 gating mechanism itself is memory-agnostic.
- 80 5. **Scaling boundaries for non-local routing:** We verify that a direct one-hop non-local
 81 edge demonstrates a flat FLOP count under distance scaling relative to multi-hop diffu-
 82 sion. However, we explicitly report that energy-gating this mechanism is outperformed
 83 by a parameter-matched, physics-free router in both floating-point operations (FLOPs) and
 84 accuracy.
- 85 6. **Capacity and epoch-budget map of the compute dial:** We map the rationing gate across
 86 capacity and training budget. The measured step-reductions ($9.9\times, 9.5\times, 6.2\times$ across
 87 kv32, kv64, kv96) are intra-CLU rationing against a full-budget CLU baseline, perfor-
 88 mance beyond kv64 is constrained by epoch-budgets rather than hard capacity limits, and

gate accuracy degrades sharply under evaluation cue noise while storage fidelity remains at 1.0.

The primary contribution of this work lies in the theoretical formalization of the certificate stack and the empirical discipline mapping its limitations, rather than claiming a general leaderboard advantage. All computational experiments were conducted on standard laptop CPUs.

2 Setup: A Conservative Memory and its Reachable Set

The governed map. A discrete state (q, p) is advanced by one dissipative velocity-Verlet step with size ε and per-step damping $\gamma \in [0, 1]$. This constitutes a kick-drift-kick integration of the separable Hamiltonian $H = T(p) + V_\theta(q)$, followed by momentum scaling $p \mapsto (1 - \gamma)p$. This update is conformally symplectic, adhering to $J^\top \Omega J = (1 - \gamma)\Omega$ and yielding $\det J = (1 - \gamma)^d$ in d -dimensions. Position-gated coupling scales this determinant to $\det J = (1 - \gamma\varphi(q'))^d$. The choice of kinetic mode dictates the bounds of T . The relativistic mode enforces a strict kinematic cap on velocity, defined as $|\dot{q}_i| < c/\sqrt{M_i}$. Conversely, the Newtonian mode leaves velocity unbounded, driven by momentum $\dot{q}_i = p_i/M_i$. To bleed excess energy toward a target E^* , a state-dependent governor $\gamma_n = s \tanh(\max(0, H - E^*))$ is applied. Complete derivations and machine-precision validations are provided in the companion theory note.

Falsifiable reachability constraints. For the governed map $\Phi_{\varepsilon, \gamma}$, the T -step reachable set originating from z_0 is defined as $R_T(z_0) = \{\Phi^n(z_0) : 1 \leq n \leq T\}$, with its position shadow denoted as $Q_T(z_0)$. To successfully access a memory state, the underlying mechanism must strictly enlarge Q_T . The theory note proves that in relativistic mode, a single drift step advances position q by at most $\varepsilon c/\sqrt{M_i}$ per coordinate. Consequently, the reachable set is strictly bounded:

$$Q_T(z_0) \subseteq C_T(q_0) := \{q : |q_i - q_{0,i}| \leq L_i, L_i = T\varepsilon c/\sqrt{M_i}\}.$$

The resulting causal box C_T is entirely energy-blind, as injected momentum asymptotically approaches, but cannot exceed, the velocity limit $c/\sqrt{M_i}$ (Appendix G.1). We define the ensuing dichotomy of failures. Kinematic reach failures occur when the target $q^* \notin C_T(q_0)$, a condition curable only by a non-local jump. Energetic escape failures occur when the target $q^* \in C_T(q_0)$ but is shielded by a potential barrier ΔV_b that the dissipating budget cannot surmount. This is resolved by any bounded energy injection that acts faster than the governor can apply braking.

In Newtonian systems, reach is energy-limited, meaning that injected energy directly translates to expanded flow reach. Our analysis deliberately focuses on the more constrained relativistic mode to enforce a strict kinematic cap. This necessitates specific resolution mechanisms because energy cannot unilaterally extend flow reach. While a squeeze mechanism can still expand reach canonically via an instantaneous displacement, it does so at an energy that grows exponentially in the rapidity ζ and is bounded by $e^{2|\zeta|}H$, establishing an energy law rather than bypassing the flow cap.

3 Verification of the Certificate Stack

To empirically validate the certificates, we employ a tightly controlled, architecturally-invariant analytic testbed. The environment consists of a symmetric double well along the reach coordinate and an $SO(2)$ sector for the latch. All results in this section are evaluated across dimensions 2 and 4, utilizing 5 random seeds, oracle channel placement, laptop-CPU execution, a sharp boundary $\gamma = 0$ rollout, and a mass band of $[4.0, 0.25]$ to yield a $16\times$ inertial contrast. This controlled setting isolates the exactness of the physical theory; learning optimal entrance placement remains outside the current scope.

3.1 Mechanism Certificates and Limitations

Escape resolution via Lorentz squeeze. To cure energetic escape failure, we apply a mass-weighted Lorentz squeeze $S_\zeta^{(M)}$ (Wales and Doye, 1997). This map is exactly symplectic ($\det = 1$) and amplifies velocity and displacement toward the saddle according to $\delta q' = \delta q \cosh \zeta + p_0 \sinh \zeta$. The injected energy is strictly bounded by $H(S_\zeta z) \leq e^{2|\zeta|}H(z)$

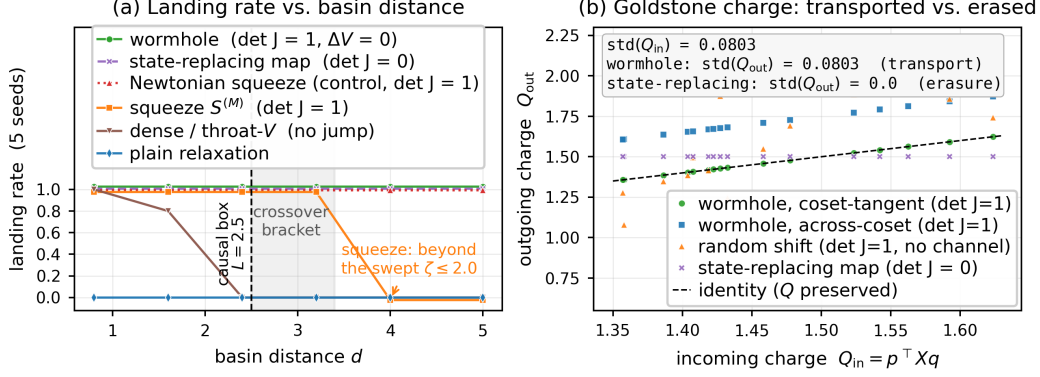


Figure 1: **Verification of the certificate stack on the designed testbed (§3).** (a) **Reach capabilities:** Landing rate as a function of basin distance d evaluated on a symmetric double-well potential (dim 2, 5 seeds, oracle placement, $L = 2.5$). Plain relaxation is escape-blocked across all distances. The Lorentz squeeze $S^{(M)}$ demonstrates a step-function response, stepping up until the required energy exceeds the swept rapidity budget $\zeta \leq 2.0$. Its theoretical reach spans $[L, L + p_0 \sinh \zeta / M_0]$ (shaded); because reach grows as $\sinh \zeta$ while the injected energy is bounded by $e^{2|\zeta|} H$, the energy required grows exponentially in rapidity ζ and therefore quadratically in the excess distance. The same mathematical bracket predicts $\zeta \approx 2.01$ to reach distance $d = 4.0$. The wormhole achieves a flat landing rate of 1.0 across all tested distances with exact volume preservation (det $J = 1$) and zero energy change. The Newtonian control confirms energy-driven flow reach, validating the relativistic cap constraint. (b) **The certificate in force:** Comparison of outgoing versus incoming Goldstone charge for 16 sample states drawn from the capture ball. The det $J = 1$ wormhole exactly transports the charge (slope 1) while preserving the spread ($\text{std}(Q_{out}) = \text{std}(Q_{in}) = 0.0803$). In contrast, an untrained det $J = 0$ state-replacing map collapses the states onto a singular charge, effectively erasing the content ($\text{std}(Q_{out}) = 0$). Note that phase volume preservation alone is insufficient; a det $J = 1$ random shift also scrambles the charge. The matched channel guarantees stable transport.

135 (Appendix G.2), and the internal governor dissipates the excess in approximately $2\zeta/\gamma_c$
 136 steps. We evaluate this on a matched quadratic potential, confirming that the energy in-
 137 jection accurately tracks the theoretical bound across varying rapidities: $(\zeta, \text{ratio}, \text{bound}) =$
 138 $(0.25, 1.13, 1.65), (0.5, 1.55, 2.72), (1.0, 3.79, 7.39), (2.0, 27.5, 54.6)$, while strictly maintaining
 139 $\det S^{(M)} = 1.000 (\pm 4 \times 10^{-6})$. We note that the $e^{2|\zeta|}$ bound is a matched-quadratic certificate;
 140 applied to a quartic well, the raw ratio can naturally exceed this limit.

141 **Reach resolution via intra-unit wormhole.** To cure kinematic reach failure, we introduce an
 142 intra-unit wormhole defined by a learned pair of loci (a, b) governed by a hard gate frozen at capture.
 143 This applies a constant canonical translation $q \mapsto q + \Delta$, $p \mapsto p$, where $\Delta = b - a$. Because this
 144 mechanism mirrors the flow of the linear Hamiltonian $W = \Delta \cdot p/\tau$, the Jacobian is the identity
 145 matrix I_{2d} , yielding $\det J = 1$ exactly. This property remains independent of Δ and the gate itself.
 146 The energy required is a discrete jump $\Delta H_{wh} = V_\theta(q + \Delta) - V_\theta(q)$ that must be tracked explicitly,
 147 where matched loci result in transport at zero energy. Evaluated on the symmetric double well with
 148 channel $-a \rightarrow +a$, the exact energy change evaluates to 0.0, successfully teleporting the state to
 149 $q = +1.000$ with $\det J = 1.000$. As a necessary design guard, if a gate varies dynamically during
 150 the jump, the volume preservation is broken by an uncompensated contraction ($\det J = 1 + \nabla g \cdot \Delta \neq$
 151 1 , measured at 2.05 in unit testing).

152 **Latch transport law.** For a protected direction maintaining a Goldstone charge $Q = p^\top X q$,
 153 where X is the broken generator, the wormhole shift exactly evaluates to $Q' - Q = p^\top X \Delta$. Because
 154 the wormhole maps a single phase point, it transports the latch rather than copying or erasing it. The
 155 charge is strictly preserved if and only if $\Delta \perp X^\top p$; otherwise, it shifts by precisely $p^\top X \Delta$. We
 156 empirically observe a zero-shift $\Delta Q = 0.0$ (predicted precision 7.5×10^{-10}) and an across-coset
 157 shift of $\Delta Q = 0.2500 = p^\top X \Delta$. A random baseline shift entirely erases the latch. The matched
 158 channel, defined as Δ being coset-tangent alongside the energy constraint, guarantees stable transit.

BIBO stability limitations. The certificate $\det J = 1$ guarantees volume preservation but does not guarantee trajectory boundedness. If an exit is placed outside the coercive connected component of the potential V_θ , the trajectory can escape to infinity, even if the jump is symplectic and the energy change is exactly zero. Evaluated on a controlled non-coercive potential ($V = \frac{1}{2}kq_0^2 - \epsilon q_0^4$), an exit at $b = 5.0$ requires zero energy ($\Delta H = 0.0$ exactly) and passes an energy-only sub-level test, yet the trajectory escapes with a growth rate proportional to the temporal horizon ($r^* \propto T$). In contrast, an admissible exit at $b = 3.0$ requires an energy change of $\Delta H = 2.88$ but remains bounded. Therefore, membership in the coercive component is the mandatory operational clause for BIBO stability. The full certificate parameters are detailed in Appendix B.

3.2 Differentiating Access Mechanisms

To systematically analyze these reachability bounds, we construct a controlled reach battery across a K -basin double well with barriers spanning across the causal box $L = 2.5$ ($T = 100$, $\varepsilon = 0.05$, $c = 1$, heavy reach coordinate $M_0 = 4.0$). The governor is parameterized such that plain relaxation is mathematically blocked from escaping (kinetic energy $0.72 < \Delta V_b = 1$). We measure the landing rates across distances $d < L = \{0.8, 1.6, 2.4\}$ and $d > L = \{3.2, 4.0, 5.0\}$.

Algorithm	$d < L$	$d > L$	Measured $\det J$	Diagnostic
Plain relaxation	0	0	$(1-\gamma)^d$	Completely escape-blocked
Squeeze $S^{(M)}$	1	$\rightarrow 0$	$1.000 \pm 4e-6$	Reach beyond C_T needs energy $\leq e^{2 \zeta }H$
Wormhole	1	1	1.0 exact	Flat access scaling; exact 0.0 energy change
Newtonian control	1	1	1.0	Confirms relativistic cap bounds flow
State-replacing map	1	1	0.0 exact	State overwritten; volume annihilated
Dense/ V -throat	$\rightarrow 0$	0	$(1-\gamma)^d$	Modulates barrier; fails global reach

The squeeze mechanism extends the reach bound to $[L, L + p_0 \sinh \zeta / M_0]$. However, because the energy scales as $e^{2|\zeta|}H$, reaching $d = 4.0$ analytically requires $\zeta \approx 2.01$, and $d = 5.0$ requires $\zeta \approx 2.64$. Consequently, reach via squeeze requires energy growing exponentially in rapidity ζ and therefore quadratically in the excess distance, whereas the wormhole’s energy requirement is fixed and independent of distance. The squeeze failing to reach the outer distances simply indicates that the energy it requires exceeds the swept rapidity budget ($\zeta \leq 2.0$), not that it is fundamentally incapable of reaching them if provided sufficient energy. Notably, the untrained state-replacing map $((q, p) \mapsto (b, p))$ demonstrates an identical landing rate to the wormhole. However, its Jacobian is mathematically block-diagonal with zero blocks, yielding $\det J = 0$ exactly, as measured by forward-mode automatic differentiation. This map is volume-annihilating and non-invertible, erasing the state completely. The Newtonian control successfully reaches past the box, confirming that the relativistic causal cap governs the primary flow limitation.

3.2.1 The Certificate’s Consequence: State Erasure vs. Transport

To evaluate the downstream consequence of the $\det J$ certificates, we track a cloud of 16 initial states drawn uniformly from the capture ball ($\text{std}(Q_{\text{in}}) = 0.0803$). It is critical to distinguish between an analytic transport map and a learned decision head. A learned decision router may select whether to utilize a non-local edge, but it transports the data through the certified $\det J = 1$ wormhole channel. The object evaluated here is a pure, untrained state-replacing map that physically overwrites the state.

Mechanism	$\det J$	ΔQ mean	ΔQ std	Predicted $p^\top X \Delta$	Max Error	$\text{std}(Q_{\text{out}})$
Wormhole (coset-tangent)	1.0000	-0.0000	0.0000	0.0000	1.2e-7	0.0803
Wormhole (across-coset)	1.0000	0.2500	0.0000	0.2500	0.0	0.0803
Random shift	1.0000	0.0972	0.2465	N/A	—	0.2793
State-replacing map	0.0000	0.0379	0.0803	N/A	—	0.0

The transport guarantee. The canonical translation of the wormhole is injective, shifting every incoming state’s charge by the exact constant $p^\top X \Delta$ (max error $\leq 1.2 \times 10^{-7}$). The topological spread is strictly preserved ($\text{std}(Q_{\text{out}}) = 0.0803$), and the pre-jump state remains exactly recoverable ($q_{\text{in}} = q_{\text{out}} - \Delta$, max err 2.2×10^{-8}).

199 **The violation.** Conversely, the state-replacing map ($\det J = 0$) acts as a volumetric black hole,
 200 mapping all 16 distinct states onto a singular bit-identical charge ($\text{std}(Q_{\text{out}}) = 0$). The coset
 201 content is irreversibly erased. This demonstrates a mechanism-level violation on a designed testbed,
 202 establishing that at $\det J = 1$, the matched canonical channel transports the stored charge, while
 203 a $\det J = 0$ state-replacing jump erases it. We explicitly note that volume preservation alone is
 204 not the latch certificate, as the random-shift baseline maintains $\det J = 1$ but still scrambles the
 205 topological spread. The matched channel preserves it.

206 4 Translating Mechanisms to Learned Memories

207 To determine the practical viability of these formalisms, we evaluate three applications on trained as-
 208 sociative memories. We transition from the analytic testbed to learned configurations, strictly track-
 209 ing the energy required by every mechanism and establishing boundaries where baseline heuristics
 210 outperform our proposed approach.

211 4.1 Memory-Agnostic Calibrated Compute-Rationing

212 We implement a self-calibrating gating mechanism on a trained CLU associative memory configured
 213 for MQAR key-value recall (vocab 256, capacity limits $\text{kv} \in \{16, 24, 32\}$, 5 seeds). We apply a
 214 Learn-then-Test (LTT; Angelopoulos et al., 2025) wrapper to determine exit thresholds based on a
 215 dynamic relaxation ladder (Geifman and El-Yaniv, 2017).

- 216 1. **Uniform transfer of calibration:** The identical calibration stack applied to a baseline
 217 modern Hopfield memory (Ramsauer et al., 2021) yields analogous improvements (raw
 218 $0.18 \rightarrow$ calibrated 0.88 , matching the CLU’s $0.43 \rightarrow 0.87$). The gating mechanism is
 219 therefore memory-agnostic; we do not assert that the CLU possesses an inherently superior
 220 energy signal.
- 221 2. **Escalatable rationing impacts low-capacity regimes:** At an under-converged 400-epoch
 222 training budget, the memory demonstrates a $4.81 \pm 0.44\times$ step-reduction at the kv16 capac-
 223 ity level compared to an always-full relaxation budget, while achieving a higher accuracy
 224 (0.894 ± 0.021 vs 0.847 ± 0.037). This efficiency gain decays as capacity demands scale,
 225 yielding $1.57 \pm 0.07\times$ at kv24, and $1.14 \pm 0.06\times$ at kv32. At full convergence (2000
 226 epochs), the gating mechanism achieves strict parity with the full-budget baseline rather
 227 than exceeding it. Therefore, the core asset of the gate is compute rationing on clean re-
 228 trieval tasks, not generalized accuracy dominance.
- 229 3. **Distribution-free certificates:** The LTT wrapper remained valid on all evaluated cells with
 230 zero violations, achieving a coverage of 0.647 ± 0.063 at a target risk of $\varepsilon = 0.05$. We note
 231 the assumption that LTT relies on exchangeability between write-time probes and deploy-
 232 ment queries, meaning the calibration operates slightly under-confident when transitioning
 233 across task domains.

234 We map the same gate across capacity and training budget on a 198-job evaluation grid (Appendix
 235 C.3). Storage fidelity converges to near-perfect levels across training epochs, indicating that the
 236 initial underperformance was an under-training artifact. Performance degradation beyond kv64
 237 is constrained by epoch-budgets rather than hard capacity limits; training models to 4000 epochs,
 238 however, causes smaller cells to over-train, degrading accuracy from 1.00 to 0.89. The measured
 239 step-reductions of the CLU gate ($9.9\times, 9.5\times, 6.2\times$ across kv32, kv64, kv96) represent intra-CLU
 240 rationing against a full-budget CLU baseline. Under evaluation cue noise ($\sigma \in \{0.3, 0.6\}$), the gate
 241 degrades sharply: at $\sigma = 0.6$ on kv32, the CLU gate yields an accuracy of 0.36, despite underlying
 242 CLU storage fidelity remaining at 1.0. The governed relaxation over-commits to the corrupted cue,
 243 and we record this noise wall as an open problem.

244 4.2 Boundary Analysis for Non-Local Routing

245 We evaluate a gated non-local wormhole designed to query distant architectural archives. We ob-
 246 serve that the direct one-hop edge maintains a flat FLOP count as distance N scales (1.18×10^8
 247 FLOP/query at $N \in \{4, 8\}$). Conversely, multi-hop chain diffusion scales poorly ($1.76 \times 10^8 \rightarrow$
 248 2.94×10^8) and demonstrates degraded distant accuracy ($0.41 \rightarrow 0.28$).

249 However, we clearly establish the boundary of this mechanism. When optimizing the routing deci-
 250 sion, energy-gating the edge is outperformed by a parameter-matched (449-parameter) physics-free
 251 router. The neural router outperforms the energy gate in both FLOPs and global accuracy (router
 252 1.000/0.948 at 8.81×10^7 FLOPs vs energy gate 0.887/0.715 at 1.18×10^8 FLOPs). Because
 253 linearly-separable per-unit key clusters allow the simple classifier to learn optimal routing trivially,
 254 relying on the relaxation energy as a routing signal is computationally sub-optimal. The direct edge
 255 remains the required transport mechanism, but energy is not the routing signal.

256 4.3 Matched-Compute Anytime Read

257 To provide a direct comparison under matched compute budgets, we evaluate the system utilizing
 258 a designed store addressed through a frozen encoder. In a regime featuring pixel-space corrup-
 259 tion where no mask oracle can be constructed, the confidence-gated anytime read ties the matched-
 260 compute feedforward floor ($+0.8 \pm 1.6$ percentage points over 6 seeds). It outperforms a 1-shot
 261 kNN floor by $+1.1 \pm 1.4$, while exceeding mechanism controls by $+4.6 \pm 2.2$ percentage points over
 262 an equal-energy random kick. We note explicitly that it ties rather than wins against the matched-
 263 compute feedforward baseline. Furthermore, where a mask oracle can be constructed, the gated read
 264 does not beat the machine-learning-optimal masked-erasure oracle in any evaluated cell, establishing
 265 a firm metric-native ceiling.

266 5 Position, Scope, and Horizon

267 We establish that on a CLU-like conservative memory, test-time compute is certified access: the
 268 phase-volume, energy and latch consequences of every access mechanism are fixed by the dynamics.
 269 A squeeze cures escape with a bounded energy injection but requires energy growing exponentially
 270 in rapidity ζ , and therefore quadratically in the excess distance, whereas a non-local wormhole
 271 achieves global reach at exact volume preservation ($\det J = 1$), an energy requirement that is
 272 fixed and independent of distance (exactly zero on the matched channel measured here), and strictly
 273 preserved latch transport. We note that while these certificates govern the physical transport of
 274 the memory state, bounding the energy change is an insufficient condition for BIBO stability; strict
 275 coercive-component screening is required. Empirically, these mechanisms facilitate scalable internal
 276 compute rationing on clean retrieval tasks for CLU-like units.

277 Furthermore, we identify an impossibility result regarding the memory read protocol. With a
 278 quadratic payload channel and a fixed read budget, a store cannot simultaneously possess exact
 279 value fidelity, amplitude-independent address hold, and amplitude-independent read latency. The
 280 implementation must sacrifice one of these properties, dictating that a faded memory mathemati-
 281 cally requires more integration steps to read.

282 Future Directions.

- 283 • **Learned entrance-steering:** Placing loci such that channel entrances naturally align with
 284 standard latent trajectories represents the primary engineering challenge for large-scale de-
 285 ployment.
- 286 • **Certifying exits on a learned potential:** The current BIBO screen relies on oracle bound-
 287 aries derived from analytic wells. Developing sub-level-set estimators for learned, non-
 288 coercive potentials V_θ is a necessary prerequisite for safe autonomous operation.
- 289 • **Re-absorption timing formalization:** The theoretical governor dissipation timescale is
 290 currently verified strictly at $\gamma = 0$. Extending these bounds across the full $\gamma > 0$ spectrum
 291 is required to fully characterize the complete relaxation trajectory.
- 292 • **Retry acceptance as a certified kernel:** Extending the certified-access framework to re-
 293 peated test-time retries requires establishing a detailed-balance kernel for the Gibbs mea-
 294 sure of the trained energy, ensuring that test-time compute acts as a Markov Chain Monte
 295 Carlo method with a strict stationarity certificate.

296 **Design rules for certified Markov kernels.** The MCMC implementation necessitates four strict
 297 design rules. First, because the state-dependent governor constitutes non-volume-preserving dissipa-
 298 tion, the actual cascade operates as Metropolis-within-annealing; we do not claim stationarity for the

governed composite. Second, because the squeeze family orbit is reducible and non-ergodic, it must be layered onto a Metropolis-Adjusted Langevin Algorithm (MALA) step (Roberts and Tweedie, 1996) to ensure adequate mixing. Third, the Langevin noise scale must be calibrated using the discrete fluctuation-dissipation relation to prevent finite-budget shadow bias. Finally, because the Gibbs measure is uniformly flat along Goldstone cosets, random walk drift will erode the stored memory register. Explicit projection orthogonal to the coset tangent is strictly required to prevent topological degradation.

References

- Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *The Annals of Applied Statistics*, 19(2), June 2025. ISSN 1932-6157. doi: 10.1214/24-AOAS1998. URL <https://projecteuclid.org/journals/annals-of-applied-statistics/volume-19/issue-2/Learn-then-test--Calibrating-predictive-algorithms-to-achieve-risk/10.1214/24-AOAS1998.full>.
- Andrea Banino, Jan Balaguer, and Charles Blundell. PonderNet: Learning to Ponder, September 2021. URL <http://arxiv.org/abs/2107.05407>. arXiv:2107.05407 [cs.LG].
- Simon Duane, A.D. Kennedy, Brian J. Pendleton, and Duncan Roweth. Hybrid Monte Carlo. *Physics Letters B*, 195(2):216–222, September 1987. ISSN 03702693. doi: 10.1016/0370-2693(87)91197-X. URL <https://linkinghub.elsevier.com/retrieve/pii/037026938791197X>.
- Yonatan Geifman and Ran El-Yaniv. Selective Classification for Deep Neural Networks, June 2017. URL <http://arxiv.org/abs/1705.08500>. arXiv:1705.08500 [cs.LG].
- Alexi Gladstone, Ganesh Nanduru, Md Mofijul Islam, Peixuan Han, Hyeonjeong Ha, Aman Chadha, Yilun Du, Heng Ji, Jundong Li, and Tariq Iqbal. Energy-Based Transformers are Scalable Learners and Thinkers, July 2025. URL <http://arxiv.org/abs/2507.02092>. arXiv:2507.02092 [cs.LG].
- Alex Graves. Adaptive Computation Time for Recurrent Neural Networks, March 2016. URL <http://arxiv.org/abs/1603.08983>. arXiv:1603.08983 [cs.NE].
- Pratik Jawahar and Maurizio Pierini. CHLU: The Causal Hamiltonian Learning Unit as a Symplectic Primitive for Deep Learning, March 2026. URL <http://arxiv.org/abs/2603.01768>. arXiv:2603.01768 [cs.LG].
- Radford M. Neal. MCMC Using Hamiltonian Dynamics. In *Handbook of Markov Chain Monte Carlo*, pages 113–162. Chapman and Hall/CRC, New York, 1 edition, May 2011. ISBN 978-0-429-13850-8. doi: 10.1201/b10905-6. URL <https://www.taylorfrancis.com/books/9780429138508/chapters/10.1201/b10905-6>.
- Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield Networks is All You Need, April 2021. URL <http://arxiv.org/abs/2008.02217>. arXiv:2008.02217 [cs.NE].
- David Raposo, Sam Ritter, Blake Richards, Timothy Lillicrap, Peter Conway Humphreys, and Adam Santoro. Mixture-of-Depths: Dynamically allocating compute in transformer-based language models, April 2024. URL <http://arxiv.org/abs/2404.02258>. arXiv:2404.02258 [cs.LG].
- Gareth O. Roberts and Richard L. Tweedie. Exponential Convergence of Langevin Distributions and Their Discrete Approximations. *Bernoulli*, 2(4):341, December 1996. ISSN 13507265. doi: 10.2307/3318418. URL <https://www.jstor.org/stable/3318418?origin=crossref>.
- Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Tran, Yi Tay, and Donald Metzler. Confident Adaptive Language Modeling. In *Advances in Neural Information Processing Systems 35*, pages 17456–17472, New Orleans, Louisiana, USA, 2022. Neural

Information Processing Systems Foundation, Inc. (NeurIPS). ISBN 978-1-7138-7108-8. doi: 10.52202/068431-1269. URL <http://www.proceedings.com/068431-1269.html>.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer, January 2017. URL <http://arxiv.org/abs/1701.06538>. arXiv:1701.06538 [cs.LG].

David J. Wales and Jonathan P. K. Doye. Global Optimization by Basin-Hopping and the Lowest Energy Structures of Lennard-Jones Clusters Containing up to 110 Atoms. *The Journal of Physical Chemistry A*, 101(28):5111–5116, July 1997. ISSN 1089-5639, 1520-5215. doi: 10.1021/jp970984n. URL <https://pubs.acs.org/doi/10.1021/jp970984n>.

A Flag-Provenance Tables

All computational results inherit the exact non-default configurations in effect.

A.1 Sec. 3 Certificate Stack Configuration

Flag	Value
JAX	0.9.0
Kinetic modes	relativistic (reach, wormhole, throat); <code>newtonian_learned</code> (Newtonian control)
Mass band	[4.0, 0.25] yielding $M_{\text{eff},0} = 4.0, 16\times$ contrast
c / rest mass / m_0	1.0 / 1.0 / 1.0
Time step dt / Horizon T	0.05 / 100 yielding box $L = T\epsilon c/\sqrt{M_0} = 2.5, v_{\text{max},0} = 0.5$
Damping γ	reach rollout $\gamma = 0$ (sharp box)
Rapidity ζ grid	[0, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0]
Basin geometry	double well along coord 0, barrier $\Delta V_b = 1.0, d \in \{0.8, 1.6, 2.4, 3.2, 4.0, 5.0\}$
Initialization	$q_0 = 0, p_0 = (1.2, 0, \dots) + 0.02\mathcal{N}$ (kinetic energy below barrier)
Landing criterion	$\min_{\text{traj}} q_0 - d < 0.4$
Seeds	reach {0, 1, 2, 3, 4}; latch <code>default_rng(0)</code>
Dimensions	2 and 4

A.2 Sec. 4.1 Calibrated Gate Configuration

Flag	Value
Task	MQAR CLU-EBM, vocab 256, capacity $k_v \in \{16, 24, 32\}$
Seeds	5 (base 42); 90 models
Write configuration	kinetic relativistic, potential mlp, hidden 128, epochs 400, lr $1e-3$, batch 16, k_{steps} 50, buffer 128, friction 0.3, temperature 0.3, input noise σ 0.05
Gate configuration	calib features <code>r_margin</code> , learned τ at write time, $p_{\text{exit}} = 0.5$; cost checkpoints 300/1200/2100/3000
Learn-then-Test	fixed-sequence, exact binomial p-values, targets $\epsilon \in \{0.05, 0.1\}$
Hopfield baseline	Platt-calibrated logit margin, identically probed

A.3 Sec. 4.2 Routing Configuration

Flag	Value
Lattice and task	distance $N \in \{4, 8\}$, embed_dim 12, vocab 128
Write configuration	kinetic relativistic, potential mlp (coercive), hidden 128, epochs 400, lr $1e-3$
Retrieval routing	dt 0.05, relax_steps 250, governor_sensitivity 0.95
Router MLP	hidden 32 (449 params), epochs 300, lr $3e-3$
Workload mixes	[[0.5, 0.5], [0.8, 0.2], [0.95, 0.05]]

365 A.4 Sec. 4.3 Regime Map Configuration

Flag	Value
Training epochs	swept across {500, 1000, 2000, 4000}
366 Kinetic and potential	relativistic / mlp (coercive)
Architecture	embed_dim 16 (CLU dim 32) / hidden 128 / vocab 256
Retrieval	relax_steps 300, governor_sensitivity 0.95, calib_r_margin
Seeds	{42, 43, 44, 45, 46}

367 A.5 Sec. 3.2.1 Latch Transport and Appendix B.2 BIBO Battery Configuration

Flag	Value
Training	none; analytic potentials, oracle channel placement
Kinetic mode	relativistic (reach, wormhole, throat, BIBO)
368 Damping γ	reach 0 (sharp box); BIBO 0.02 (bounded arms must settle)
BIBO potential	$k = 1.0$, $\epsilon = 0.02$ yielding $x_b = 3.5355$, barrier $V_b = 3.125$
BIBO parameters	exits $b \in \{1.0, 2.0, 3.0, 3.6, 4.0, 5.0\}$; horizon $T = 2000$
Latch transport	16 incoming states, ball radius $0.3 < \rho = 0.35$
Seeds	reach/BIBO {0, 1, 2, 3, 4}; latch cloud default_rng(0)

369 We note that the two primary configuration axes influencing comparative performance are damping
370 and training epochs. The calibrated gate evaluation utilizes $\gamma = 0.3$ and 400 epochs, capturing
371 an under-converged regime. In contrast, the regime mapping sweeps up to 4000 epochs to isolate
372 converged behavior.

373 B The Access-Certificate Table and BIBO Battery

374 B.1 Table 1: Certificate per Mechanism

Mechanism	Energy Injected	Volume det J	Governor Re-absorption	Latch Impact
Wormhole (translation)	$\Delta V = V(b) - V(a)$, discrete jump	1 exactly	$\approx (1/\gamma_c) \ln(1 + \Delta V/E^*)$	Transport via shift $p^\top X \Delta$
376 Wormhole (throat)	standing wave, no jump	$(1 - \gamma)^d$ exact	continuous	adds $\mu^2 \propto A/\ell^2$
Squeeze $S^{(M)}$	$\leq (e^{2 \zeta } - 1)H$	1 exactly	$\approx 2\zeta/\gamma_c$	preserved

377 Both jump mechanisms mandate evaluation alongside a coercive potential V_θ . Because deep convo-
378 lutional potentials are architecturally non-coercive out-of-unit, exits must be strictly constrained to
379 lie inside a coercive sub-level component.

380 B.2 The BIBO Battery

381 We evaluate the bounding capabilities on a controlled non-coercive well where the connected com-
382 ponent spans $|q_0| < x_b = 3.536$. We assess the maximum excursion radius r^* at horizons T and
383 $2T$.

Algorithm (r^* at $2T$)	$b=1.0$	2.0	3.0	3.6	4.0	5.0
Wormhole with certificate	1.01	2.01	3.01	0.09	0.09	0.09
Wormhole, ablation	1.01	2.01	3.01	102.13	103.43	104.83
384 State-replacing map	1.01	2.01	3.01	102.13	103.43	104.83
Escape rate, certified	0	0	0	0	0	0
Escape rate, blind	0	0	0	1.0	1.0	1.0
Wormhole certificate logic	ADMIT	ADMIT	ADMIT	REJECT	REJECT	REJECT
Energy sub-level test	admit	admit	admit	reject	admit	admit

385 The escaping arms present a linear growth rate $r^*(2T)/r^*(T) \approx 1.9$, demonstrating genuinely
386 unbounded trajectories constrained only by the terminal velocity cap. The critical evaluation oc-
387 curs at exit $b = 5.0$, where the energy change is exactly zero, causing an energy-only sub-level

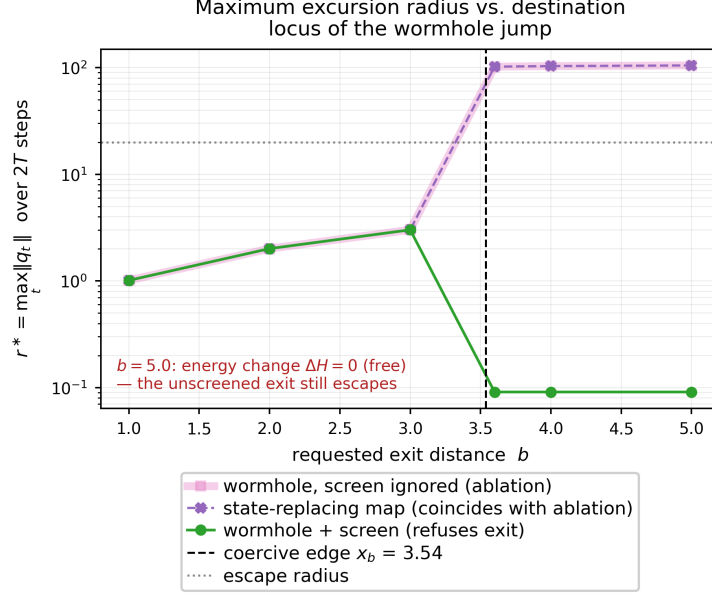


Figure 2: **The BIBO battery (App. B.2, verification grade).** Maximum excursion radius $r^* = \max_t \|q_t\|$ at horizon $2T$, against the requested exit locus b . Exits inside the coercive component stay bounded for every arm. Beyond the coercive edge $x_b = 3.54$ the unscreened arms grow as $r^* \propto T$, while the screened wormhole holds at $r^* = 0.09$ because it *refuses* the jump rather than making it safe. The state-replacing map coincides exactly with the screen-ignored ablation, so what provides boundedness is the screen and not the jump. Note $b = 5.0$: the energy change is exactly zero and the exit still escapes.

test to erroneously admit the jump, resulting in immediate escape. This conclusively demonstrates that $\det J = 1$ combined with a zero energy change is insufficient for BIBO stability; coercive-component membership is the requisite property. Furthermore, the blind wormhole and state-replacing maps coincide exactly on all failures, proving that the certificate screening mechanism, not the jump itself, provides the bounding guarantee.

C Detailed Evaluation Grids

C.1 Reach Battery Full Grid

Evaluated	across	5	seeds	with	causal	box	L	=	2.5.
Algorithm	$d=0.8$	1.6	2.4	3.2	4.0	5.0			
Plain relaxation	0	0	0	0	0	0			
Squeeze $S^{(M)}$	1	1	1	1	0	0			
Wormhole	1	1	1	1	1	1			
Newtonian control	1	1	1	1	1	1			
State-replacing map	1	1	1	1	1	1			
Dense/V-throat	1	0.8	0	0	0	0			

The squeeze requires more energy than the swept rapidity grid provides at $d = 4.0$. The mathematical bracket accurately predicts that reaching $d = 4.0$ requires a rapidity of $\zeta \geq 2.0105$, corresponding to an energy bound growing as $e^{2|\zeta|}$ in the rapidity. The state-replacing map is an untrained analytic constant map with $\det J = 0$ exactly, distinct from the learned decision head evaluated in the routing section.

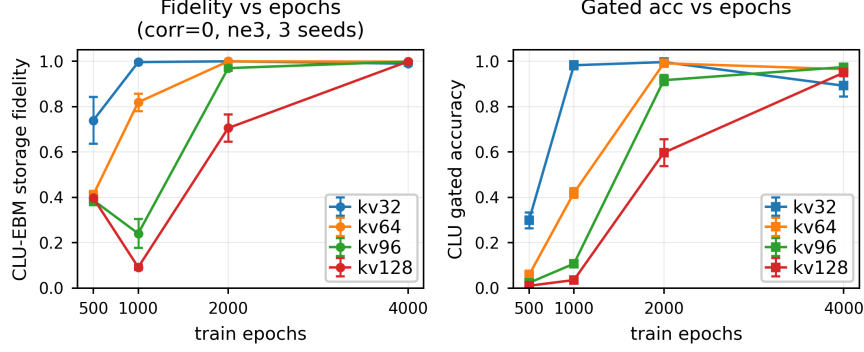


Figure 3: **Epoch-scaling frontier (App. C.3, evidence grade; $n = 3$ seeds).** CLU-EBM storage fidelity and gated accuracy against training budget for $\text{kv} \in \{32, 64, 96, 128\}$. Required epochs scale with capacity rather than saturating at a fixed ceiling, and kv32 over-trains between 2000 and 4000 epochs. Note this panel is $n = 3$, not the 8 pooled seeds of the capacity axis below.

C.2 Routing Full Grid

Evaluated across 5 seeds for local and distant accuracy (L/D) and floating-point operations per query.

Algorithm	$N=4$ Acc (L/D)	$N=4$ FLOP	$N=8$ Acc (L/D)	$N=8$ FLOP
Local only	0.500 (1.00/0.00)	5.88e7	0.500 (1.00/0.00)	5.88e7
Gated energy	0.887 ± 0.139 (0.90/0.88)	1.18e8	0.715 ± 0.172 (0.82/0.61)	1.18e8
Dense network	0.500 (0.00/1.00)	1.18e8	0.448 ± 0.070 (0.00/0.90)	1.18e8
Diffusion chain	0.652 ± 0.159 (0.90/0.41)	1.76e8	0.548 ± 0.138 (0.82/0.28)	2.94e8
Physics-free router	1.000 ± 0.000 (1.00/1.00)	8.81e7	0.948 ± 0.070 (1.00/0.90)	8.81e7

C.3 Regime Map: Capacity Axis and Epoch Frontier

Evaluated with zero correlation across 8 pooled seeds. The grid characterizes the capacity and epoch-budget axes of the compute dial of §4.1.

Cell (N,kv)	Epochs	CLU Fidelity	Gate Acc	Intra-CLU Step-Reduction
N128/kv32	500	0.76 ± 0.09	0.31 ± 0.04	1.2×
N128/kv32	2000	1.00 ± 0.00	1.00 ± 0.00	9.9×
N256/kv64	500	0.43 ± 0.05	0.06 ± 0.02	1.0×
N256/kv64	2000	1.00 ± 0.00	0.99 ± 0.00	9.5×
N384/kv96	500	0.40 ± 0.03	0.02 ± 0.01	1.0×
N384/kv96	2000	0.97 ± 0.01	0.91 ± 0.02	6.2×

The epoch-scaling frontier confirms that required epochs scale with capacity. While kv32 saturates by 1000 epochs, kv96 requires 4000 epochs to achieve an accuracy of 0.975. We explicitly observe over-training artifacts where smaller cells degrade when trained to excessive horizons.

D Documented Negative Results

We explicitly record several primary negative findings to demarcate the boundaries of our mechanisms. First, the squeeze retries recover no performance at a single-unit scale when selecting among already reachable attractors, demonstrating that the squeeze addresses kinematic reach failure rather than local selection. Second, the compute dial does not confer robustness to corrupted cues (§4.1): gate accuracy degrades sharply under evaluation cue noise while storage fidelity remains perfect, indicating that the governed relaxation over-commits to the corrupted cue rather than failing to store the pattern. Noise-aware thresholds, longer relaxation budgets and denoising initialization are open work. Finally, we confirm that the raw energy signal adds negligible predictive value over a standard readout margin, establishing that our framework relies on the calibrated rationing mechanism itself rather than an inherently superior physics signal.

E Analytic Verification Protocols

All verifications were conducted using standard numerical precision implementations in Python on CPU architectures. The reach causal cap was validated to a relative error of 2.0×10^{-12} . The Jacobian preservation was validated to 1.000, with state-dependent variations correctly tracking the expected analytical deviations to 2.2×10^{-16} . The Goldstone latch transit error remained bounded below 1.8×10^{-15} . Finally, the governor re-absorption mechanics accurately tracked the theoretical exponential injections across all swept rapidities.

F Derivations for Certified Markov Kernels

To secure a stationarity certificate for test-time retries, we derive the four operating constraints for the proposed Markov kernel. First, because the symmetric Lorentz squeeze is exactly volume-preserving ($\det J = 1$), it requires no Jacobian correction in the acceptance ratio. A retry proposing a sign-symmetrized squeeze and accepting via $\min(1, e^{-\Delta H/T})$ forms a detailed-balance kernel for $e^{-H/T}$ (Duane et al., 1987). However, because the state-dependent governor constitutes non-volume-preserving dissipation, the deployed sequence is effectively Metropolis-within-annealing, converging to a colder maximum-a-posteriori state rather than acting as a pure Gibbs sampler.

Second, the squeeze family orbit represents a single hyperbola, rendering it non-ergodic. Ergodicity strictly necessitates momentum refreshment, requiring the squeeze to be layered with a Metropolis-Adjusted Langevin Algorithm step to ensure proper mixing across the state space (Neal, 2011).

Third, the Langevin noise scale must be calibrated using the discrete fluctuation-dissipation relation, defined as $\sigma_i^* = \sqrt{M_{\text{eff},i} T \gamma (2 - \gamma)}$ (Roberts and Tweedie, 1996). Applying the correct calibration reduces the shadow bias finite-budget error from 0.0995 to 0.0065. In relativistic kinetic modes, the target distribution follows Maxwell-Jüttner statistics; therefore, fixed-covariance Gaussian refreshes are invalid, and stationarity is solely maintained by the Metropolis-Hastings corrections.

Finally, because the Gibbs measure is uniformly flat along Goldstone cosets, any coset-tangent proposal has an energy differential of zero and is accepted uniformly. This dynamic generates an unbiased random walk that diffuses and erodes the stored register at a rate proportional to the proposal scale variance. Explicit projection orthogonal to the coset tangent successfully quenches this drift, proving that without projection, even a certified MCMC retry mechanism will progressively erase stored memory contents.

G Derivations for the Reach and Injection Certificates

This appendix derives the two closed forms the main text relies on — the causal box of §2 and the bounded-injection certificate $H(S_\zeta z) \leq e^{2|\zeta|} H(z)$ of §3.1 — together with the hard-gate Jacobian quoted in §3.1. Everything follows from the governed map of §2. Throughout, $z = (q, p) \in \mathbb{R}^d \times \mathbb{R}^d$, and in relativistic mode the reference architecture’s kinetic energy is

$$T(p) = \sqrt{c^2 p^\top M^{-1} p + m_0^2 c^4},$$

with $M = \text{diag}(M_1, \dots, M_d) \succ 0$ the learned inertial mass and $m_0 > 0$ the rest mass (values in the flag table).

G.1 The causal box

Claim. One drift sub-step advances each coordinate by strictly less than $\varepsilon c / \sqrt{M_i}$, for every momentum p and every potential V_θ ; consequently $Q_T(z_0) \subseteq C_T(q_0)$ with $L_i = T \varepsilon c / \sqrt{M_i}$.

Derivation. The drift sub-step of the kick-drift-kick map moves only the position, $q \mapsto q + \varepsilon \nabla_p T(p)$, at whatever momentum is current at that sub-step. Differentiating T ,

$$\frac{\partial T}{\partial p_i} = \frac{c^2 p_i}{M_i T(p)}, \quad T(p) = \left(c^2 \sum_j p_j^2 / M_j + m_0^2 c^4 \right)^{1/2} > \frac{c |p_i|}{\sqrt{M_i}},$$

the last inequality strict for every p because $m_0 > 0$ and the remaining kinetic terms are non-negative. Hence

$$|\dot{q}_i| = \left| \frac{\partial T}{\partial p_i} \right| = \frac{c^2 |p_i|}{M_i T(p)} < \frac{c^2 |p_i|}{M_i} \cdot \frac{\sqrt{M_i}}{c |p_i|} = \frac{c}{\sqrt{M_i}},$$

so a single drift obeys $|q'_i - q_i| < \varepsilon c / \sqrt{M_i}$. The two kick sub-steps and the damping $p \mapsto (1 - \gamma)p$ act on momentum only and move no position, so the same bound holds for the full step, uniformly in θ , γ , and the state. Summing over $n \leq T$ steps gives $|q_{n,i} - q_{0,i}| < n \varepsilon c / \sqrt{M_i} \leq L_i$, i.e. $Q_T(z_0) \subseteq C_T(q_0)$.

Energy-blindness. The bound involves only (ε, c, M_i) : neither V_θ nor the magnitude of p enters. Injected momentum drives $|\dot{q}_i|$ toward the cap from below, with relative deficit $1 - |\dot{q}_i| \sqrt{M_i} / c \approx m_0^2 c^2 M_i / (2p_i^2)$ as $|p_i| \rightarrow \infty$ (momentum along coordinate i) — the saturation stated in §2. With the §3 configuration ($T = 100$, $\varepsilon = 0.05$, $c = 1$, $M_0 = 4.0$) the reach coordinate has $L = 2.5$ and $v_{\max,0} = 0.5$, the values used throughout §3. In Newtonian mode $\nabla_p T = M^{-1}p$ is unbounded, no such box exists, and reach is energy-limited, which is why the Newtonian control of §3.2 crosses the box.

G.2 The bounded-injection certificate

The map. On the targeted coordinate pair (q_i, p_i) , pass to mass-weighted coordinates $u = \sqrt{M_i} q_i$, $v = p_i / \sqrt{M_i}$ and let the mass-weighted squeeze $S_\zeta^{(M)}$ act as the hyperbolic rotation

$$u' = u \cosh \zeta + v \sinh \zeta, \quad v' = v \cosh \zeta + u \sinh \zeta,$$

identity on all other coordinates. In the original variables, $q'_i = q_i \cosh \zeta + (p_i / M_i) \sinh \zeta$ and $p'_i = p_i \cosh \zeta + M_i q_i \sinh \zeta$. The map is linear with determinant $\cosh^2 \zeta - \sinh^2 \zeta = 1$ on the active pair and 1 elsewhere, hence exactly symplectic: this is the $\det S^{(M)} = 1$ certificate of §3.1.

Displacement law and the reach bracket. Applied at the launch state of the reach battery ($q = 0$, momentum p_0 along the heavy coordinate), the squeeze produces the instantaneous displacement $q'_0 = (p_0 / M_0) \sinh \zeta$; the subsequent governed flow advances by at most the box half-width L (Appendix G.1). The squeeze-extended reach is therefore contained in the bracket $[L, L + p_0 \sinh \zeta / M_0]$ of Figure 1 and §3.2. Landing at distance d within the tolerance 0.4 of the flag table requires $(p_0 / M_0) \sinh \zeta \geq d - 0.4 - L$, i.e.

$$\zeta \geq \operatorname{arcsinh}((d - 0.4 - L) M_0 / p_0),$$

which at $L = 2.5$, $p_0 = 1.2$, $M_0 = 4.0$ evaluates to $\zeta \geq 2.0105$ for $d = 4.0$ and $\zeta \geq 2.64$ for $d = 5.0$ — the thresholds quoted in §3.2. Both exceed the swept budget $\zeta \leq 2.0$, which is why the squeeze row of the reach battery drops to zero at $d \geq 4.0$ while $d \leq 3.2$ remains covered.

The energy bound. The certificate is a matched-quadratic statement: on the active pair the quadratic Hamiltonian is isotropic in the same mass-weighted coordinates, $H_q = \frac{1}{2} \kappa (u^2 + v^2)$ with $\kappa > 0$. Under the squeeze,

$$u'^2 + v'^2 = (u^2 + v^2) \cosh 2\zeta + 2uv \sinh 2\zeta \leq (u^2 + v^2) (\cosh 2\zeta + |\sinh 2\zeta|) = (u^2 + v^2) e^{2|\zeta|},$$

using $2|uv| \leq u^2 + v^2$. Hence $H(S_\zeta z) \leq e^{2|\zeta|} H(z)$, with equality exactly on the diagonal $u = \pm v$ (sign matched to that of ζ), and the injected energy obeys $\Delta H \leq (e^{2|\zeta|} - 1)H$, the form quoted in the certificate table. Spectator coordinates only tighten the bound: they contribute the same non-negative energy to both sides and pull the ratio toward one. Two consequences used in the text: at the launch state $u = 0$ the ratio equals $\cosh 2\zeta$ exactly, so the measured ratios of §3.1 must lie between $\cosh 2\zeta$ and the bound $e^{2|\zeta|}$, as they do; and the bound is attained only on the diagonal, so generic states sit strictly below it.

Exponential in rapidity, quadratic in excess distance. Writing $\delta = (p_0 / M_0) \sinh \zeta$ for the displacement beyond L , the launch-state ratio is

$$\cosh 2\zeta = 1 + 2 \sinh^2 \zeta = 1 + 2 (M_0 \delta / p_0)^2,$$

exactly quadratic in δ , while the certificate $e^{2|\zeta|} = (\cosh \zeta + |\sinh \zeta|)^2$ is exponential in the rapidity and likewise quadratic in δ at large ζ . This is the scaling stated in the abstract and §5; the energy is *not* exponential in the distance itself.

507 **Scope.** The inequality is proved for the matched quadratic H . On a quartic well the potential term
 508 transforms with higher powers of $\cosh \zeta$, so the raw ratio can exceed $e^{2|\zeta|}$, exactly as reported in
 509 §3.1.

510 **G.3 The hard-gate Jacobian**

511 **Claim.** If the gate is allowed to vary during the jump, $q \mapsto q + g(q)\Delta$, $p \mapsto p$, then $\det J =$
 512 $1 + \nabla g(q) \cdot \Delta$; freezing the gate at capture (g constant during the jump) restores $\det J = 1$ exactly.

513 **Derivation.** The Jacobian is block-triangular with momentum block I_d and position block the
 514 rank-one update $\partial q' / \partial q = I_d + \Delta \nabla g(q)^\top$, so by the matrix determinant lemma

$$\det J = \det(I_d + \Delta \nabla g(q)^\top) = 1 + \nabla g(q)^\top \Delta.$$

515 The defect is first order in the gate's variation along the jump, so no smallness of Δ removes it; this
 516 is the design guard quoted in §3.1 (unit-test value 2.05).